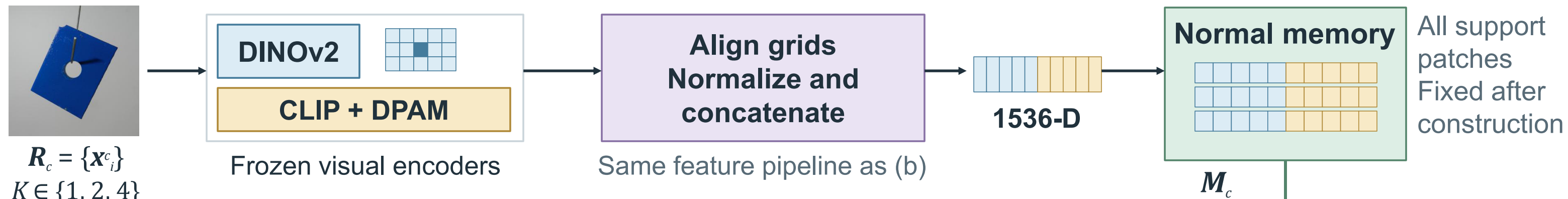


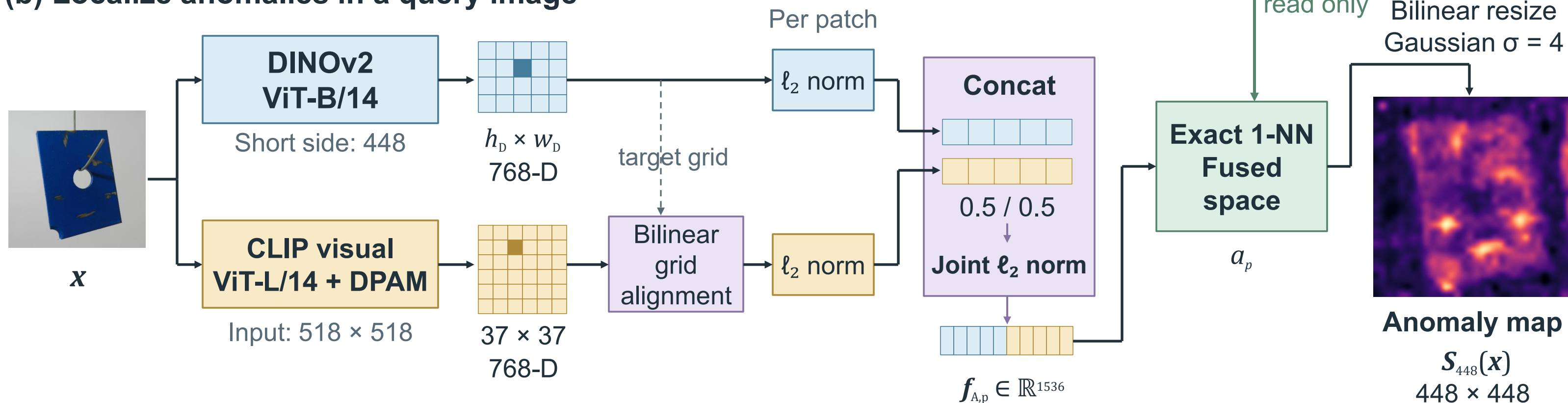
DCFnet Dual-Encoder Calibrated Fusion Network

Normal-only few-shot industrial anomaly localization

(a) Build the normal reference memory



(b) Localize anomalies in a query image



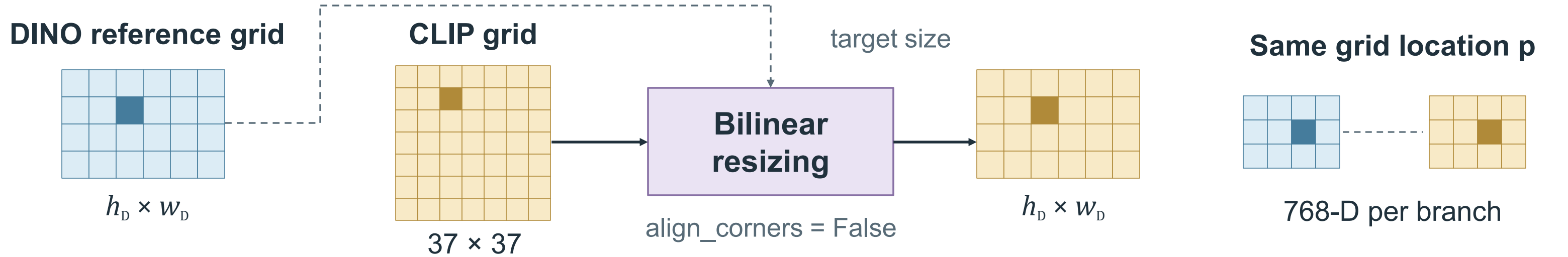
$$a_p = \frac{1}{2} \min_{m \in M_c} \|f_{A,p}(x) - m\|_2^2$$

$$s_{\text{img}}(x) = \max_u S_{448,u}(x)$$

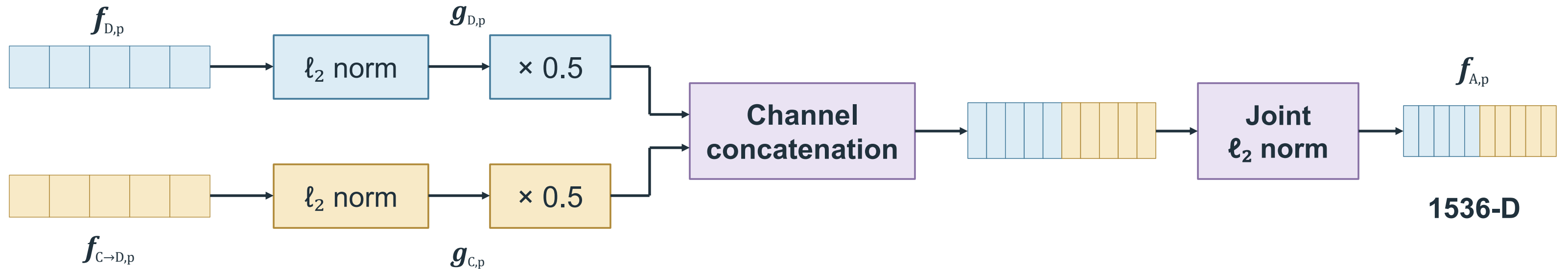
Fig. 2 Spatial calibration and fixed feature fusion

Common spatial sampling, normalized branch magnitudes, and a joint descriptor

(a) Align the CLIP grid to the DINO lattice



(b) Normalize, scale and concatenate each patch pair

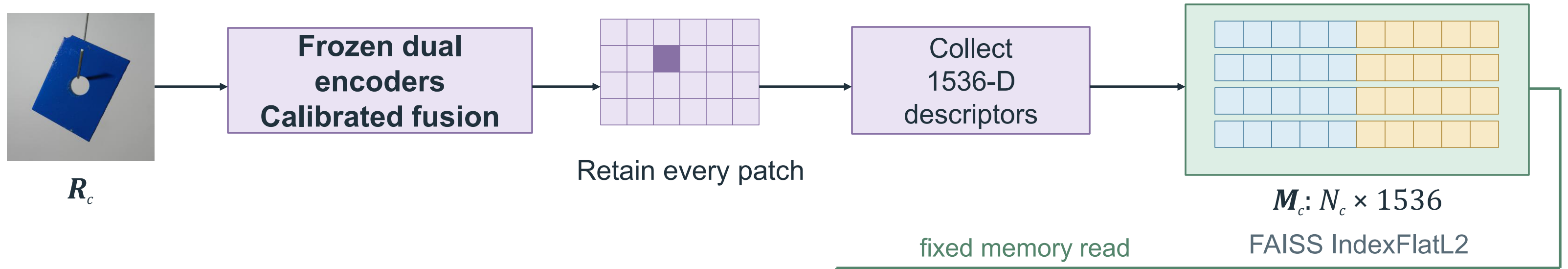


$$f_{A,p} = \text{Norm}([0.5 \mathbf{g}_{D,p}; 0.5 \mathbf{g}_{C,p}]) = [\mathbf{g}_{D,p}; \mathbf{g}_{C,p}] / \sqrt{2}$$

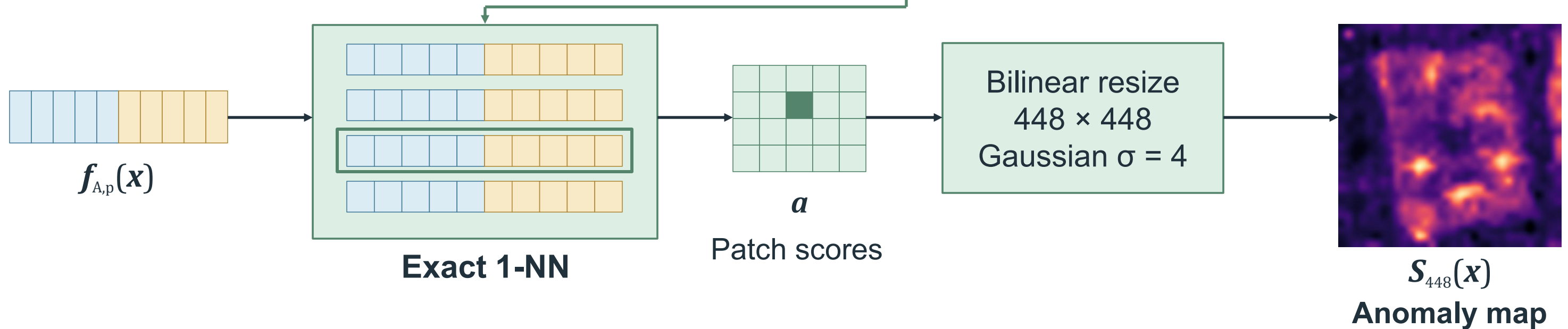
Fig. 3 Normal reference memory and anomaly scoring

All normal support patches form the fixed memory used for query retrieval

(a) Collect the normal support descriptors



(b) Find the nearest support patch in feature space

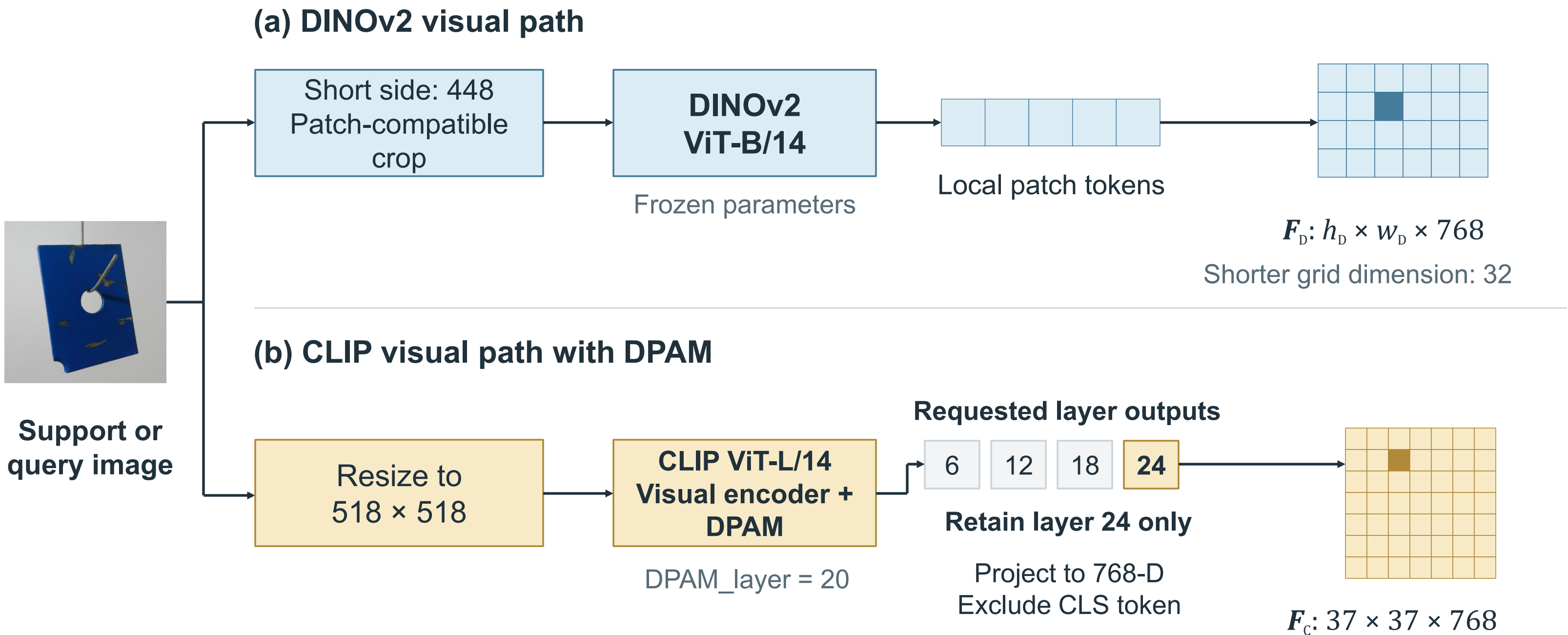


$$a_p = \frac{1}{2} \min_{m \in M_c} \|f_{A,p}(x) - m\|_2^2$$

$$s_{\text{img}}(x) = \max_u S_{448,u}(x)$$

Fig. S1 Frozen dual-encoder visual representation

Independent image preprocessing produces two 768-dimensional patch representations

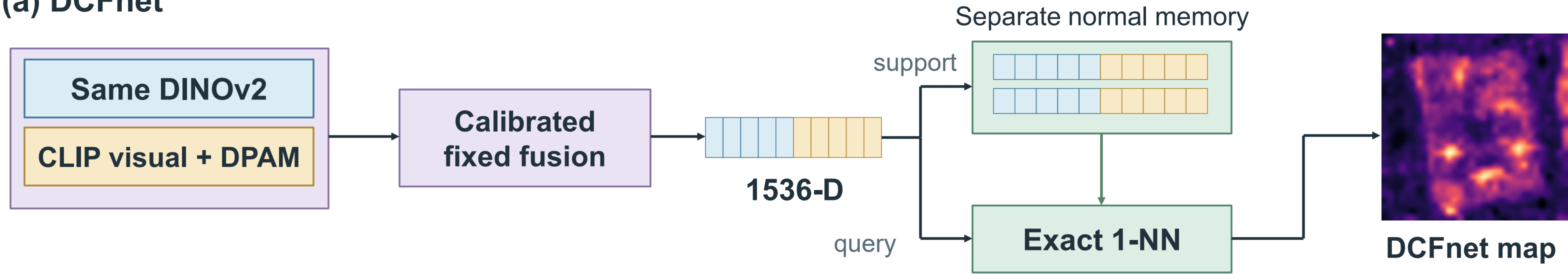


Frozen visual features for both paths. No text prompts or text encoder at inference.

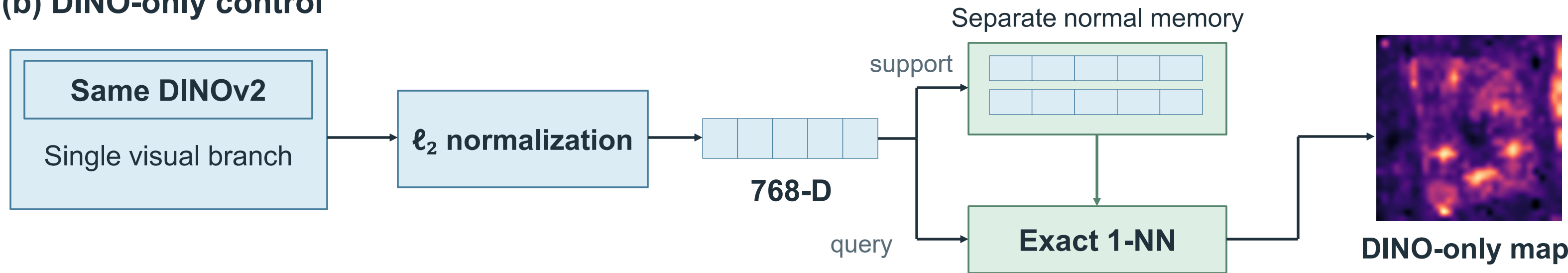
Fig. S2 Matched DINO-only control

Identical support identities, query images and downstream scoring protocol

(a) DCFnet



(b) DINO-only control



Same scoring and post-processing: squared ℓ_2 / 2, resize, Gaussian $\sigma = 4$, spatial maximum

Paired maps: one existing MPDD example, seed 0, K = 1